

WHY DO BIRDS ADD GREEN NEST MATERIAL?

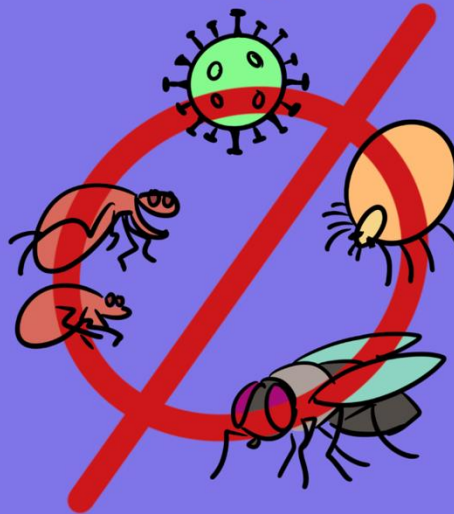
SYSTEMATIC REVIEW AND META-ANALYSIS

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Bird nests are fascinating complex structures
.... crucial to survival and reproduction ...





Why did we do a meta-analysis?

1. Does the addition of green nest material provide any (adaptive) benefits across bird species that perform this behaviour?

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2. Is there any evidence for the proposed mechanism of the benefits?

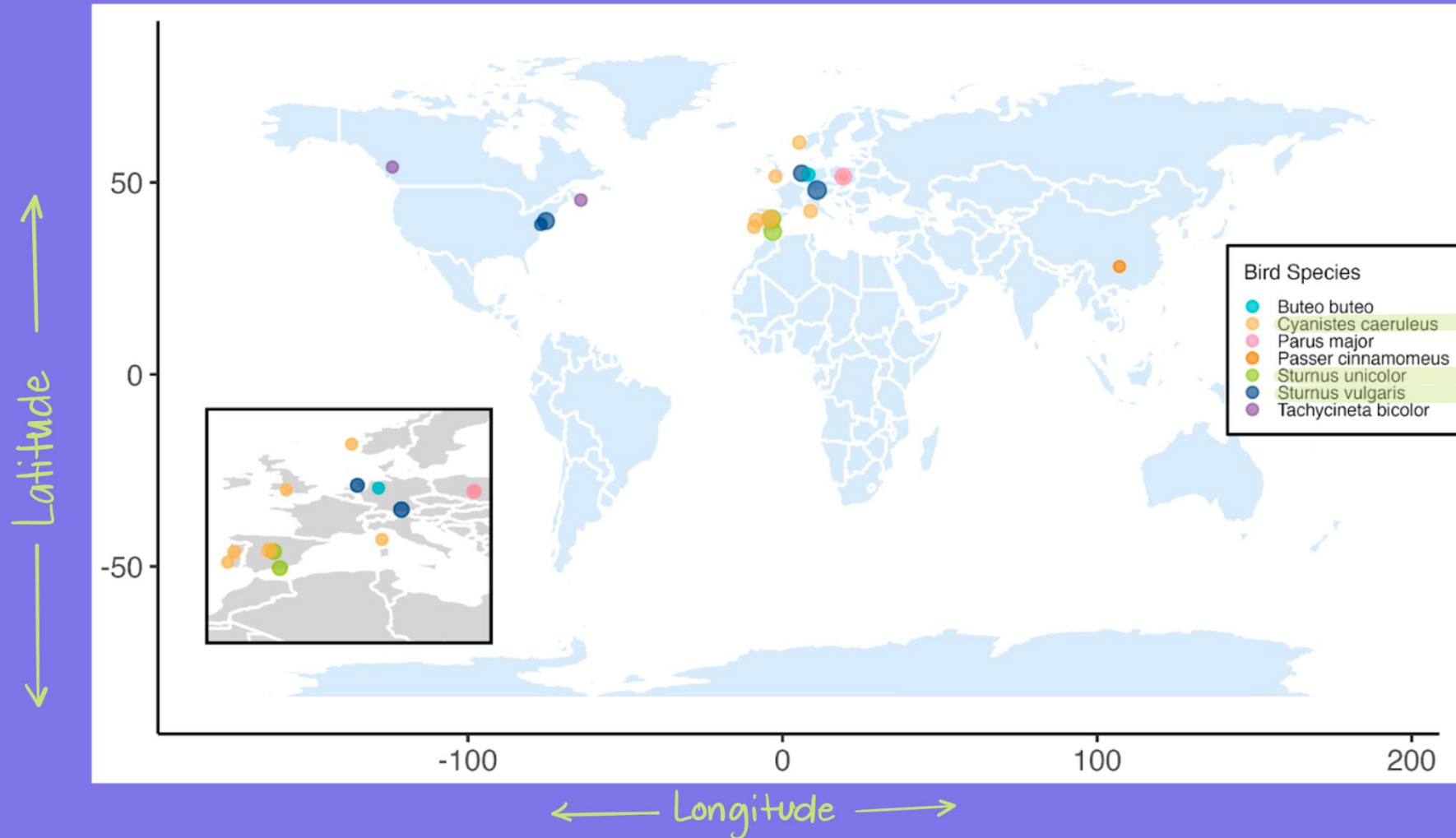
Why did we do a meta-analysis?

1. Does the addition of green nest material provide any (adaptive) benefits across bird species that perform this behaviour?
2. Is there any evidence for the proposed mechanism of the benefits?
3. If there is variation in this observed effect (heterogeneity), can we explain some of the drivers?

Overview of the literature

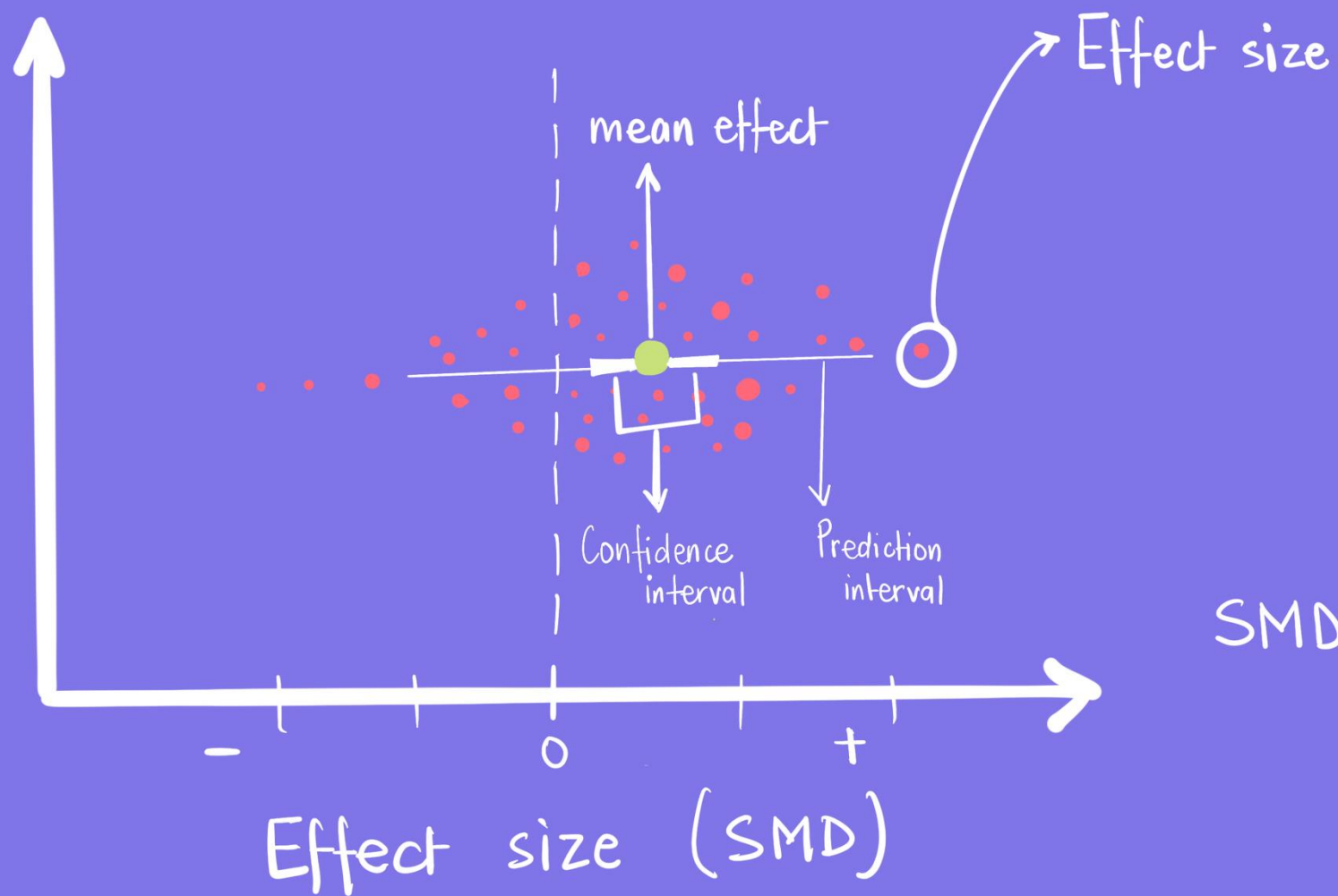
N = 26 published
+ 2 unpublished
studies

k = 274 effect sizes



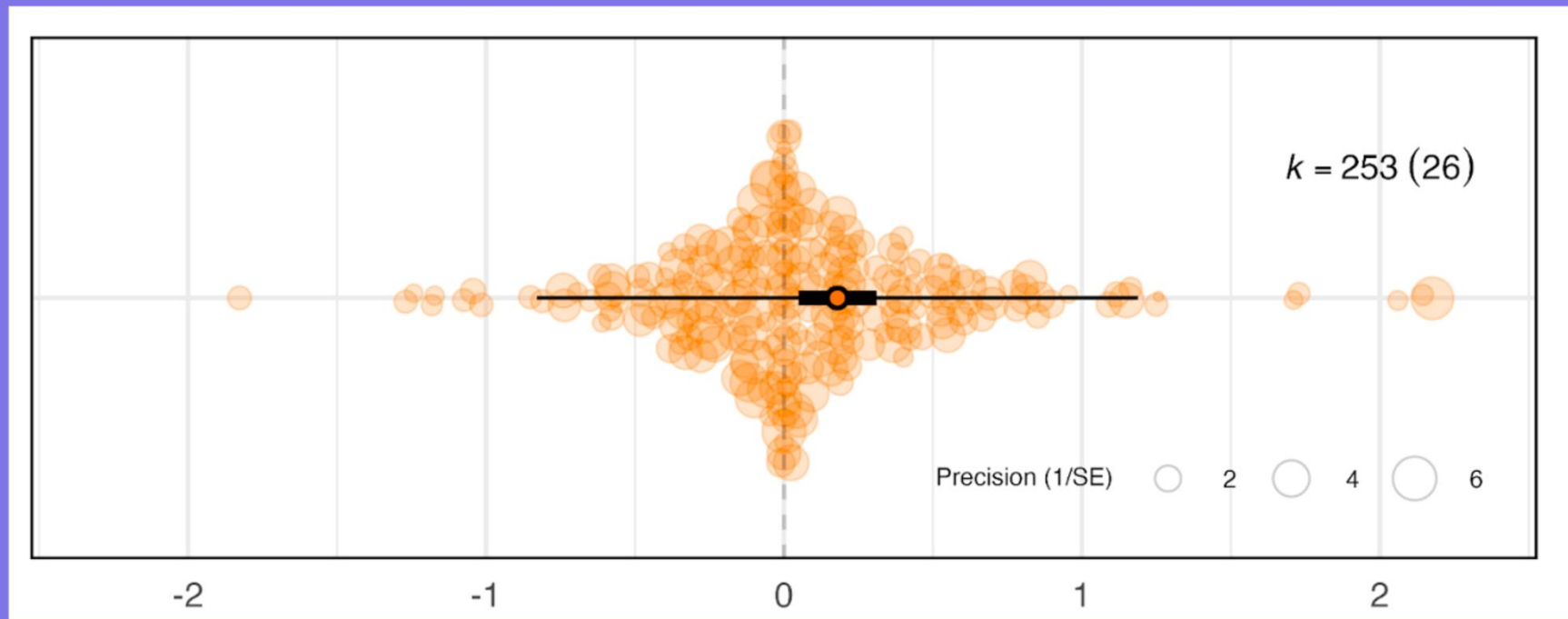
81% of effect
sizes came from it

How to read the plots?



$$SMD = \frac{\mu_{\text{Treatment}} - \mu_{\text{control}}}{SD_{\text{pooled}}}$$

Overall Effect $\rightarrow$ Small, positive, consistent



Effect size (SMD)

Evidence for mechanism → Unclear, non-conclusive

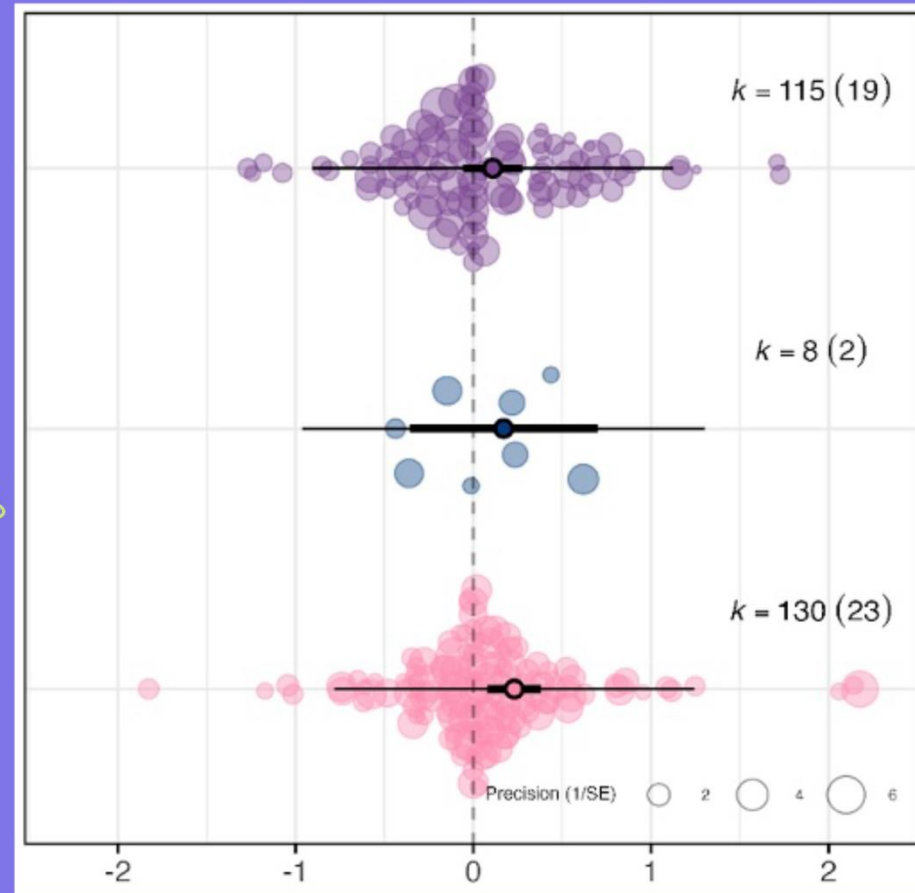


Parental care
hypothesis



Courtship
hypothesis

Both/Unclear
mechanism



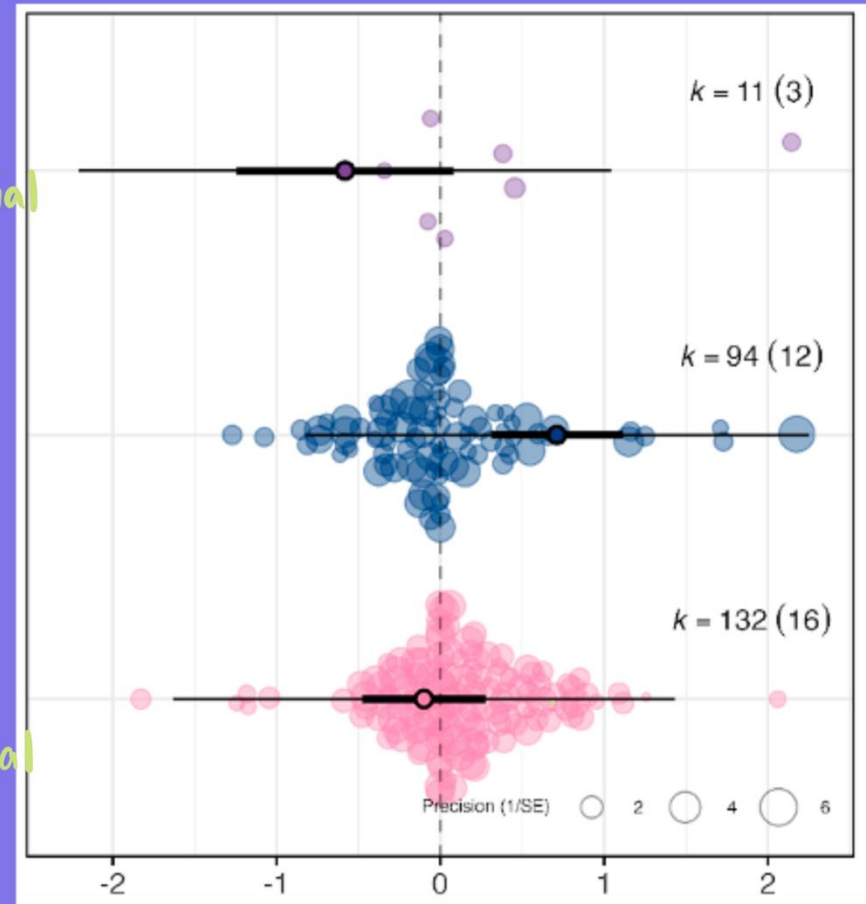
Effect size (SMD)

Driver of heterogeneity → Experimental design

No material vs.
non-aromatic material

No material vs.
Aromatic material

Non-aromatic vs.
Aromatic material



Effect size (SMD)

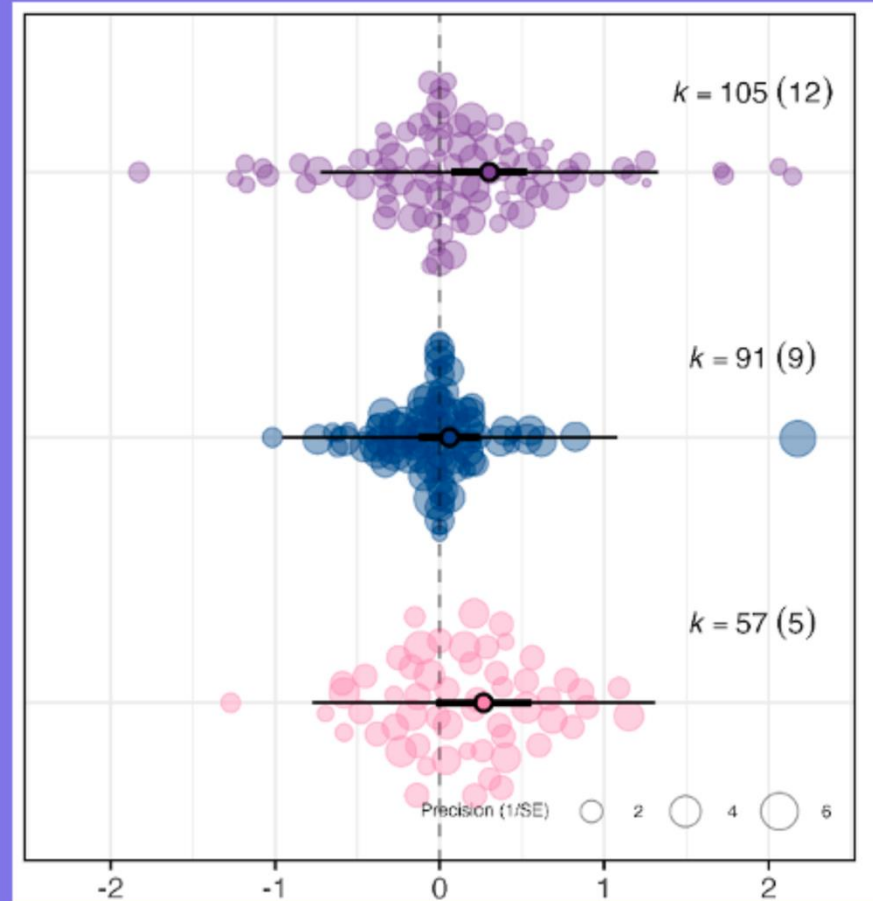
Aromatic property
of the material
might not be the
driving factor

Driver of heterogeneity → Experimental design

Continuous addition
of material

Before egg laying

After egg laying



Green material added
"before egg-laying only,"
had no adaptive benefit

Take Home

- Green material added to nests has a positive fitness outcome across bird species ...
 - Evidence is limited to Blue Tits, European and Spottless starlings
 - Might benefit from expanding species and geographical location

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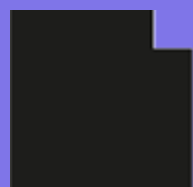
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- Green material added to nests has a positive fitness outcome across bird species ...
 - Evidence is limited to Blue Tits, European and Spotless starlings
 - Might benefit from expanding species and geographical location
- No evidence of mechanism of this effect
 - Experiments to disentangle effect, reconsider fitness outcome...
- Aromatic material might not be the driving factor ...
 - Other mechanisms or functions - micro-climate, structural integrity



Alfredo Sánchez-Tójar



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Colleagues... Tuba, Julio, Meinolf

Researches who do these
experiments ...

Gladalski et. al. 2020, 2024

Pires et. al. 2020

Polo et. al. 2004, 2007, 2015

Lopez-Rull and Gill 2009

Shulter and Campbell 2007

Brouwer and Komdeur 2002

Ruiz-Castellano et. al. 2016

Tomás et al. 2012, 2013

Soler et. al. 2017

Dawson 2004

Favth et. al. 1991

Scott-Bauman et. al. 2022

Gwinner et. al. 2000, 2005, 2018

Mennerat et. al. 2008, 2009

Clark and Mason 1988

Garrido-Bautista 2023

Yang et. al. 2020

Master Student (Mennerat Lab)

